

Fig. 2: Debug disassembly window (Source: <https://microchip.wdfiles.com>)

available in C-callable structure format.

Graphics and cryptographic libraries. MPLAB system allows the drawing to enable faster updates and reduces the need to screen painting. The draw tool window can be zoomed. Other facilities include copy/paste and add new resources, support for GPU-enabled MCUs and a display manager to quickly enable the use of displays.

While graphics library supports drawing window, cryptographic library helps to perform functions such as encryption, decryption, hash-

ing, compression and authentication within an embedded application.

System services for managed shared resources include:

Clock. This library provides control of the oscillator module on microcontrollers during operation nodes.

Console. It is required to debug or re-route errors from PIC32 microcontroller to terminal program. Thus it helps in diverting traffic from the console to the middle-layer application.

Debug. It allows users to fix bugs at global level.

Device control. This service initiates the developer with a simple application programming interface to configure and control device-specific functions.

Interrupt. It initialises the processor's interrupt controllers and manages them.

Messaging. This feature provides intra-process and inter-process communication by sending and receiving simple messages.

Many functions like DMA, ports, reset, timer and WDT are also listed in system services.

MPLAB Harmony Version 2.0: One step ahead

WYSIWYG engine. Ranging from graphic asset utilisation management to image transformation, and from compression to editing, the What You See Is What You Get (WYSIWYG) engine helps in more accurate GUI representation.

MISRA-C:2012 compliance. MPLAB version 2.0 now complies with MISRA-C:2012 standards due to upgrades in peripheral code library.

Board Support Package (BSP) manager. Updates to the software framework also include additional tools and support such as Board Support Package (BSP) Manager and updated pin-manager.

Other updates. A new Graphics Composer suite and enhanced tools support easy GUI creation.

To conclude

MPLAB Harmony is an open source freeware designed to work with MPLAB XC32 compilers and PIC32 solutions. Its updated version includes libraries of peripherals, drivers and system services to support application development. Code operability, faster time to market, improved compatibility and quick support make MPLAB a complete software for end-product development. **EFY**

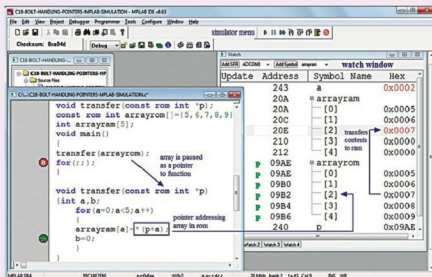


Fig. 3: Use of pointer in MPLAB (Source: www.puntotofantante.net)

Please visit <http://www.microchip.com/mplab/mplab-harmony> to download MPLAB and know more about its features. This software is not included in the DVD.